SWAN CHEMICALS

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# Lead Evaluator

***Minimum Viable Product (MVP)***

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# Problem Statement

## Common Practice

- Manual lead verification
- Time heavy duty
- No standardization
- Decision varies
- Risk of inconsistency

## Solution

- **Automatic** verification
- Takes seconds
- **SOP** established
- **Learns** gradually
- **Consistent** results

# Dataset Overview

*Synthetic Data*

**3,000** Dummy Companies

**5** Lead Buckets

**3** Districts

**8** Industries

**10** Locations

**4** Company Sizes

**3** Digital Signals

Required Amount

Urgency Days

**25** Columns

Payment Methods

*Machine Learning Model*

# CatBoost Classifier

# Model Chosen

1. Multinomial **Logistic Regression**
2. **Decision Tree** Classifier
3. **LightGBM** Classifier
4. **HistGradientBoosting** Classifier
5. **CatBoost** Classifier



# Model Performance

**78.15%**
Macro F1

**4.22%**
Overfitting

**79.00%**
Accuracy

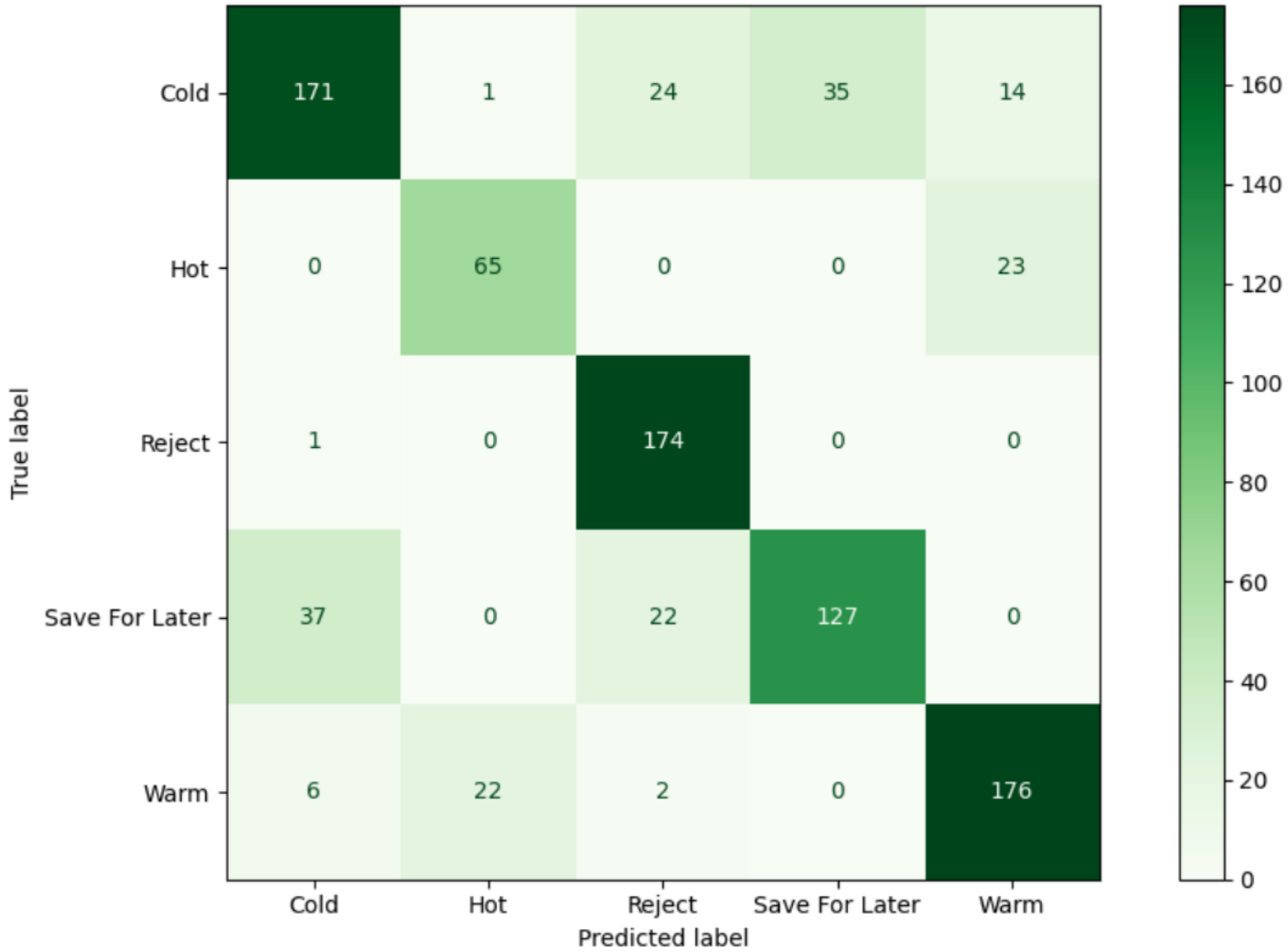
**82.13%**
Hot & Warm Recall

**99.16%**
Qual-vs-Not AUC

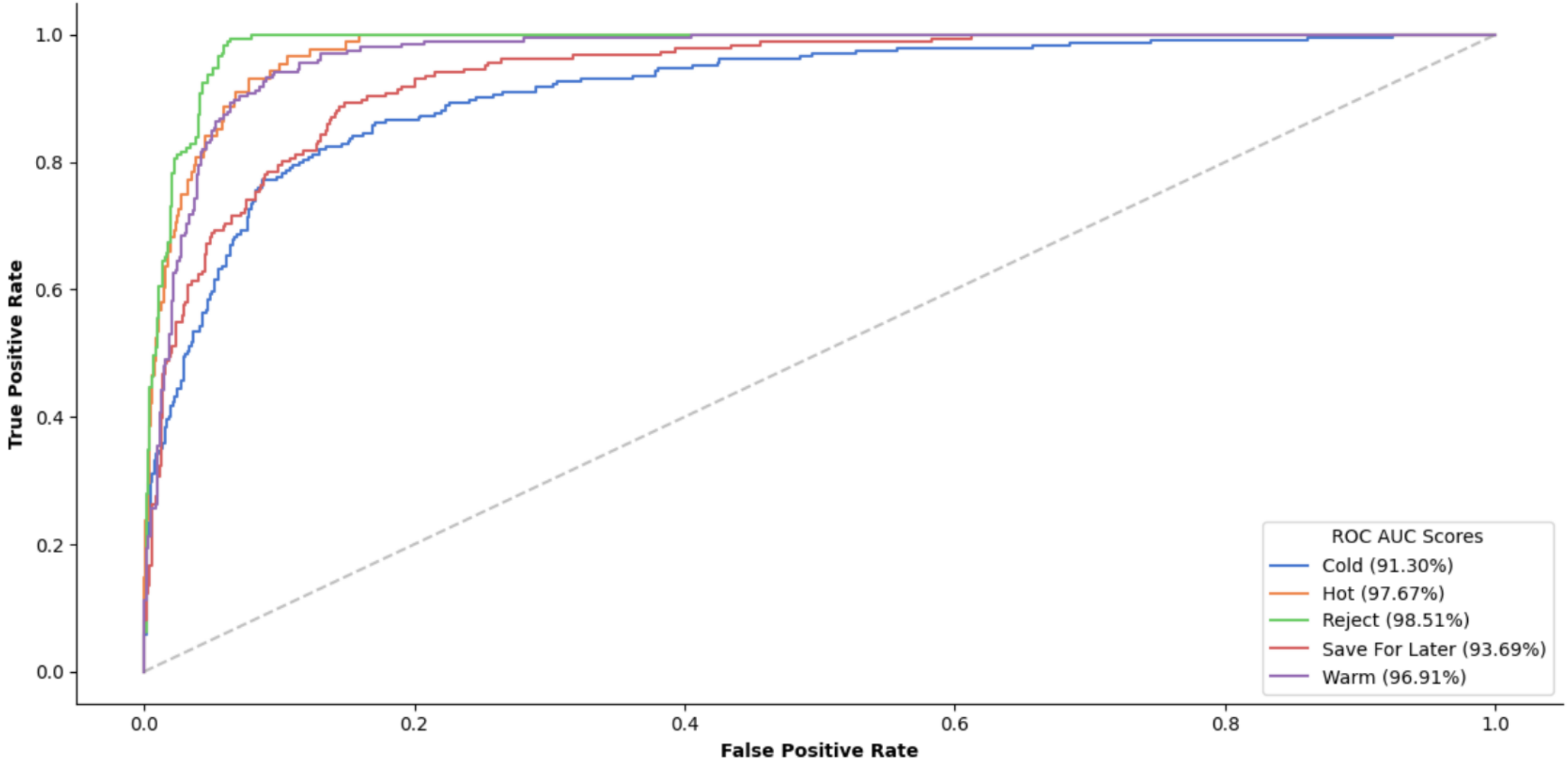
**95.10%**
Weighted ROC AUC

**99.62%**
Macro ROC AUC

# Confusion Matrix



# One-vs-Rest ROC Curve



# Project Architecture

## *model_development.ipynb*

### Development Workspace

- Clean Data
- Test Models
- Tune Hyperparameter
- Measure Performance
- Select Final Model
- Save Artifacts

## *pipeline.py*

### Production-Style Workflow

- Load Saved Artifacts
- Replicate Clean Workflow for Testing
- Create Function for Testing

## *score_new_leads.py*

### Testing Script

- Takes CSV as input
- Runs Function from ***pipeline.py***
- Produces Output

# What the Outputs Look Like?

# Terminal Command

```
python3 score_new_leads.py "datasets/leads_test.csv"
```

~/Library/Mobile Documents/com~apple~CloudDocs/7th Semester/ml/deliverables/04_lead_evaluator git:(main) 2 files changed, 4 insertions(+), 24 deletions(-) (0.051s)

```
clear
```

~/Library/Mobile Documents/com~apple~CloudDocs/7th Semester/ml/deliverables/04_lead_evaluator git:(main) 2 files changed, 4 insertions(+), 24 deletions(-) (

```
python3 score_new_leads.py "datasets/leads_test.csv"
```

| Company Name | Actual Lead Bucket | Predicted Lead Bucket | Status | Actual Lead Score | Predicted Lead Score | Score Difference | Action |
|---|---|---|---|---|---|---|---|
| Company_1802 | Reject | Reject | Match | 28 | 0.4 | 27.6 | Deprioritize |
| Company_1191 | Reject | Reject | Match | 13 | 0.2 | 12.8 | Deprioritize |
| Company_1818 | Reject | Reject | Match | 15 | 0.3 | 14.7 | Deprioritize |
| Company_0252 | Warm | Warm | Match | 69 | 82.9 | 13.9 | Prioritize (Sales) |
| Company_2506 | Cold | Cold | Match | 47 | 2.5 | 44.5 | Deprioritize |
| Company_1118 | Save For Later | Save For Later | Match | 48 | 0.9 | 47.1 | Deprioritize |
| Company_1412 | Warm | Warm | Match | 83 | 97.1 | 14.1 | Prioritize (Sales) |
| Company_2114 | Save For Later | Save For Later | Match | 36 | 1.0 | 35.0 | Deprioritize |
| Company_0409 | Save For Later | Reject | Mismatch | 43 | 0.7 | 42.3 | Deprioritize |
| Company_2580 | Hot | Hot | Match | 90 | 98.3 | 8.3 | Prioritize (Sales) |
| Company_2255 | Save For Later | Cold | Mismatch | 40 | 3.7 | 36.3 | Deprioritize |
| Company_0053 | Cold | Save For Later | Mismatch | 51 | 0.9 | 50.1 | Deprioritize |
| Company_0645 | Hot | Hot | Match | 85 | 95.2 | 10.2 | Prioritize (Sales) |
| Company_1764 | Reject | Reject | Match | 8 | 0.8 | 7.2 | Deprioritize |
| Company_2777 | Warm | Hot | Mismatch | 80 | 99.0 | 19.0 | Prioritize (Sales) |
| Company_0901 | Reject | Reject | Match | 26 | 0.6 | 25.4 | Deprioritize |
| Company_1074 | Cold | Cold | Match | 50 | 2.4 | 47.6 | Deprioritize |
| Company_0473 | Cold | Cold | Match | 35 | 2.8 | 32.2 | Deprioritize |
| Company_2739 | Cold | Cold | Match | 49 | 2.9 | 46.1 | Deprioritize |
| Company_2547 | Warm | Warm | Match | 64 | 58.0 | 6.0 | Nurture (Follow-up) |

```
Bucket Accuracy: 80% (16/20)
Average Score Difference: 27.0 points
```

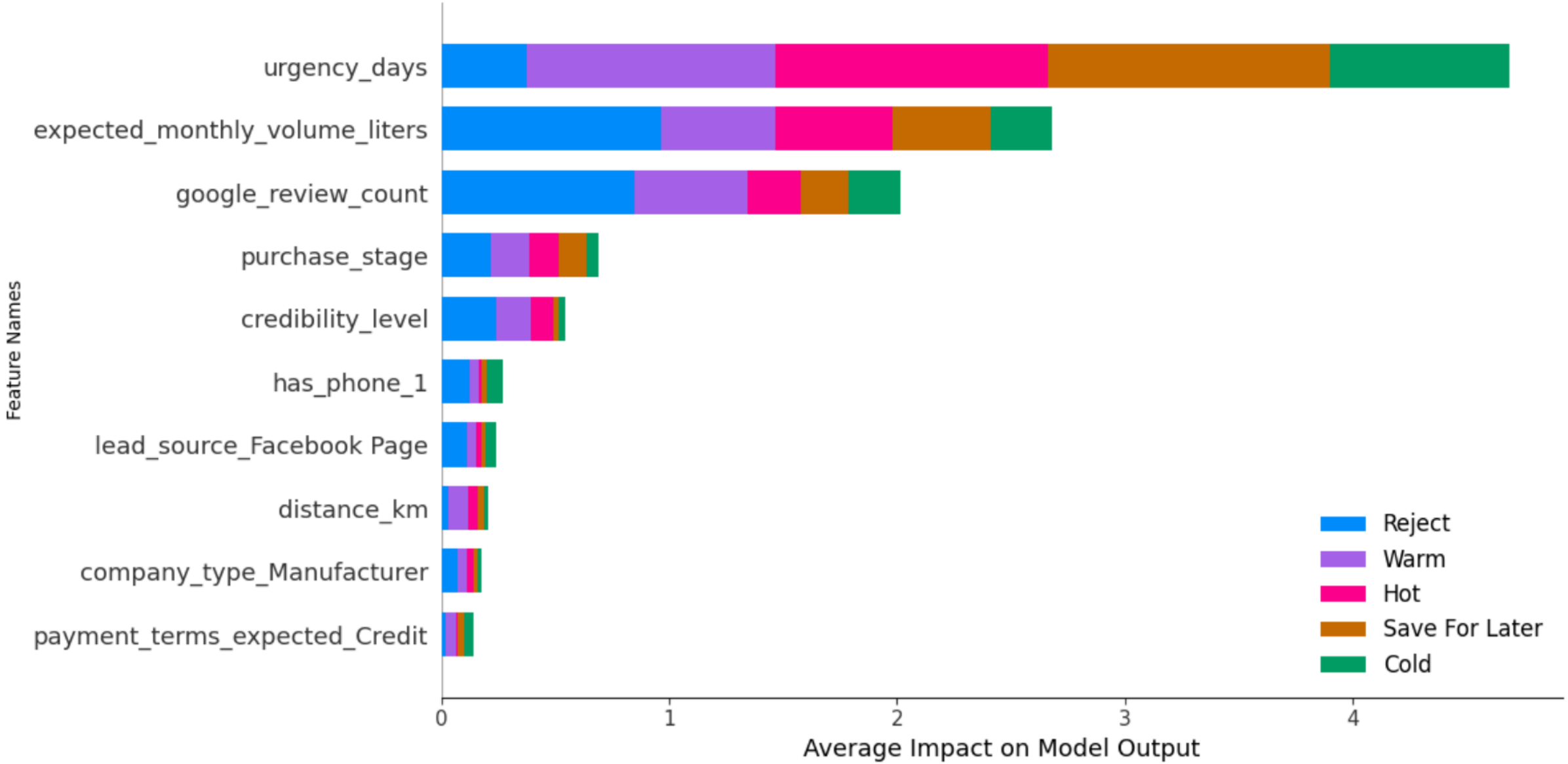
…ter/ml/deliverables/04_lead_evaluator  main  3 • +4 -24

Warp anything e.g. Help me debug why my Python tests are failing in CI

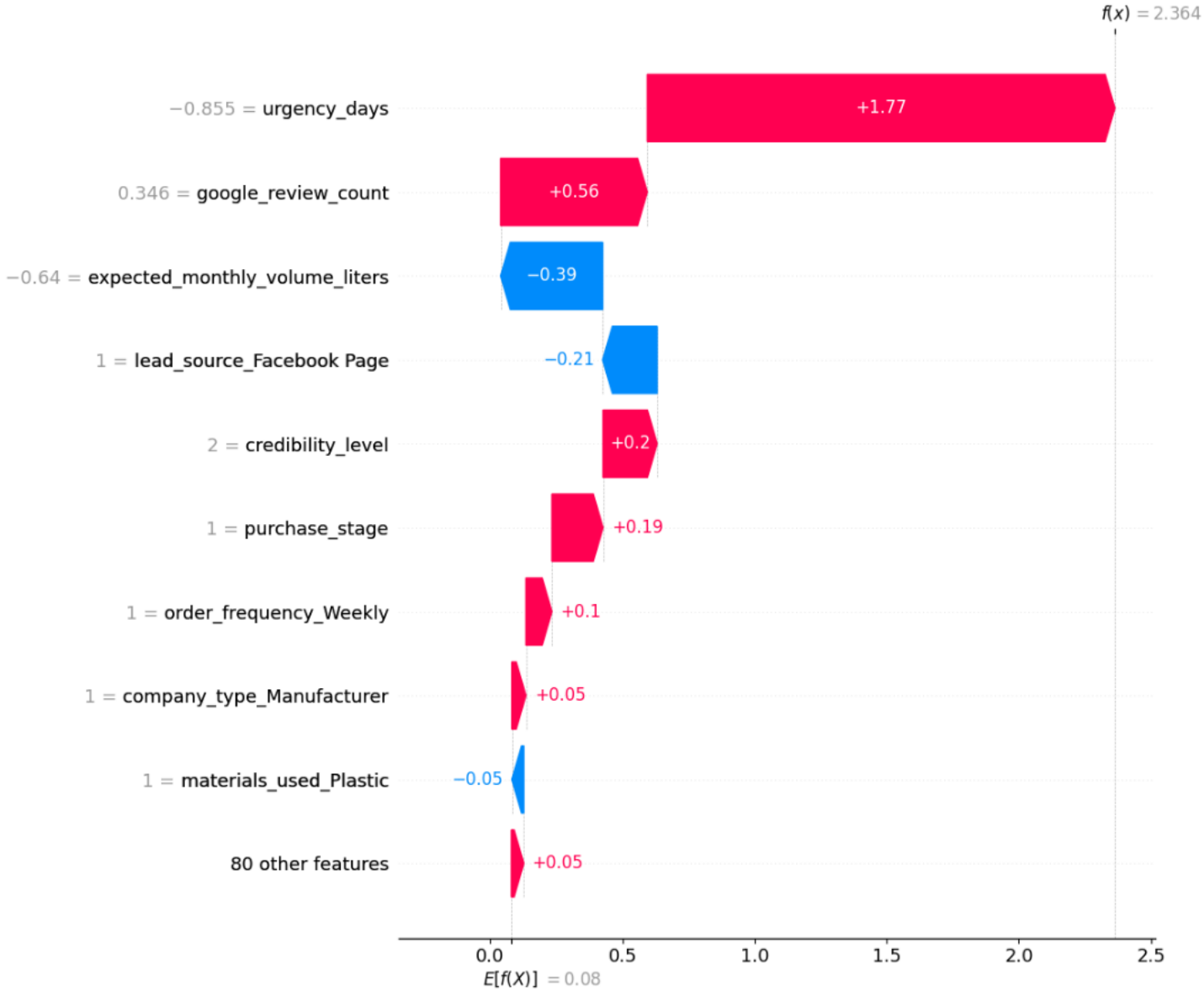
⌘↵ new /agent conversation with `python3 score_new_leads....` attached

# Feature Insights & Business Interpretation

# Top 10 Features by Average Impact on Predictions



# Feature Contribution Breakdown (Warm)



# Next Stages

Real-World Pilot Testing

Continuous Model Refinement

System Integration

Expand to Revenue Forecasting

# Thank You